

Problem Statement:

You are required to implement One-Sample and Two-Sample t-tests in Python using the following datasets:

Dataset 1 (Energy Bar Weights – grams):

[49.5, 50.2, 48.7, 51.1, 49.9, 50.5, 49.8, 50.3, 49.7, 50.6,
50.1, 49.4, 50.0, 49.6, 50.8, 49.3, 50.4, 50.7, 49.2, 50.9]

The company claims the average weight is 50 grams. Test this claim using a one-sample t-test at 5% significance.

Dataset 2 (Student Scores – Teaching Methods A & B):

method_A = [78, 85, 82, 90, 76, 88, 84, 79, 91, 77]

method_B = [72, 81, 79, 74, 69, 83, 78, 75, 80, 71]

Test whether there is a significant difference between the two teaching methods using a two-sample t-test at 5% significance.

Tasks:

- a) State the null and alternative hypotheses for both cases.
- b) Check normality of the datasets using visualization (histogram/QQ plot).
- c) Perform the appropriate t-tests using Python (scipy.stats).
- d) Report the t-statistic and p-value for both tests.

e) Interpret the results in plain English (reject/fail to reject null hypothesis).